

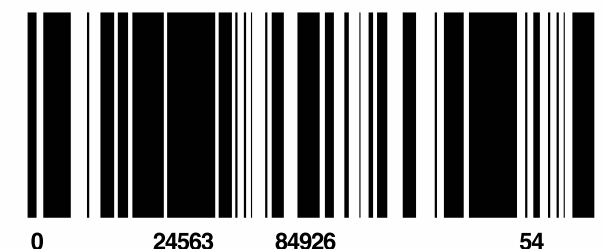
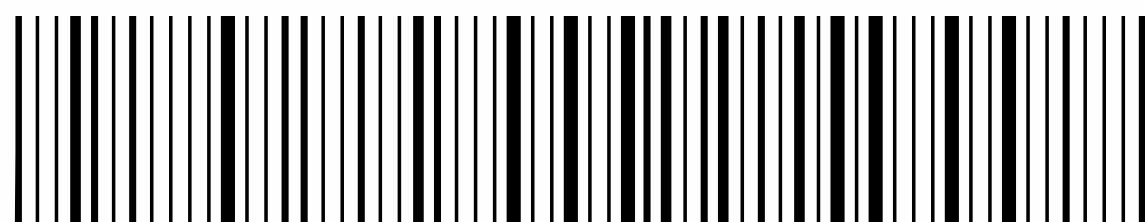
LET'S GO



**HOLIDAY!!**

**HKG → PVG**

**BOARDING 18:50**  
**FLIGHT 19:20**





Dear Mr Tanawongsuwan Fahtai



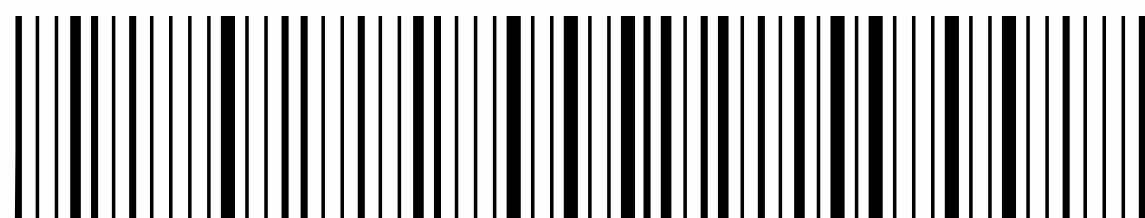
## Flight status update

### CX 362

Your flight **CX 362** has been delayed due to the late arrival of your operating aircraft. We apologise for this disruption to your travel plans. Thank you for your patience and understanding.

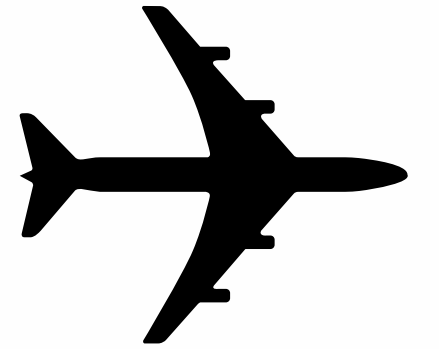


**ARRIVING 18:50**  
**~~FLIGHT 19:20~~**

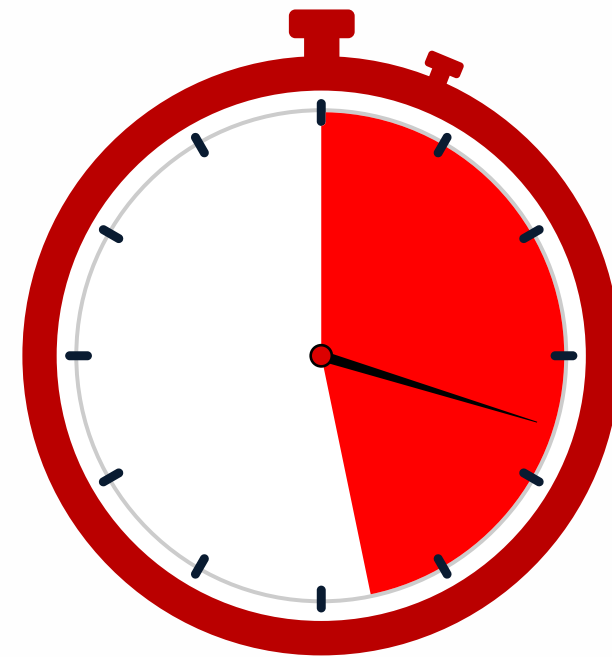


**SIKES FLIGHT 20:20+++**

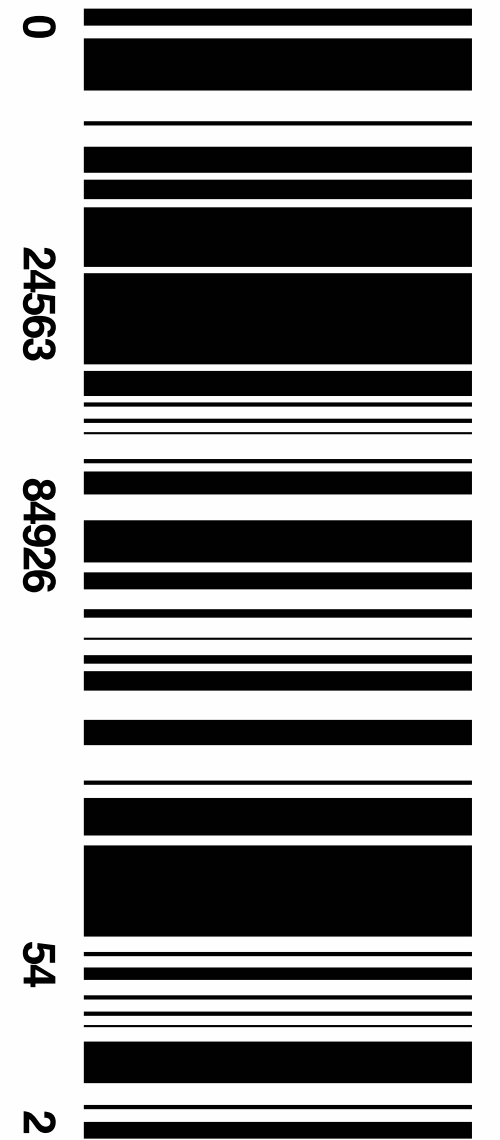
# FLIGHT DELAYS



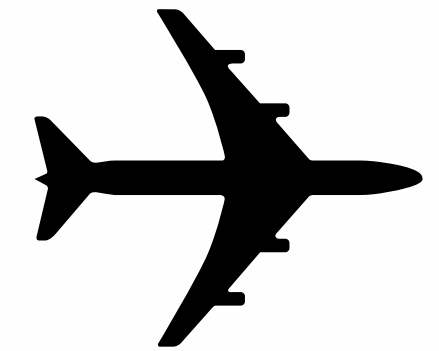
## PATTERNS? AVOIDABLE?



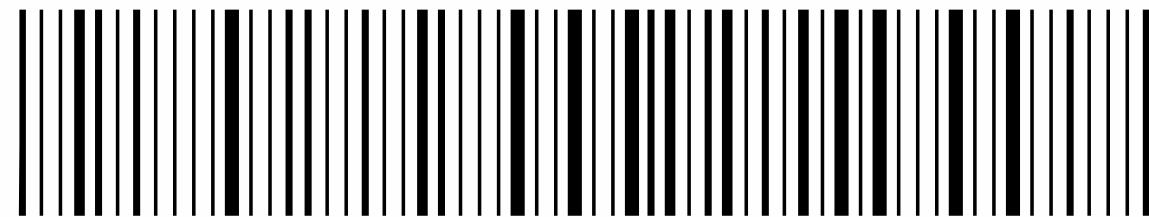
**Fahtai Tanawongsuwan**  
**(3036538194)**



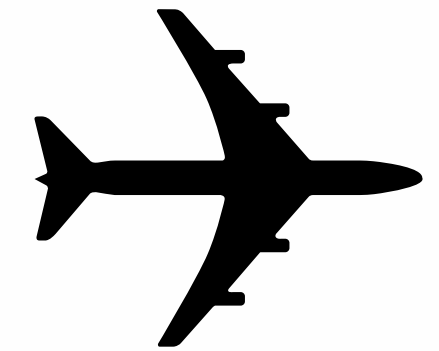
# FLIGHT DELAYS:



- **Is my delayed flight just unlucky?**
- **Are there any factors that consistently correlates with delays**
- **Can we choose flights to avoid delays**

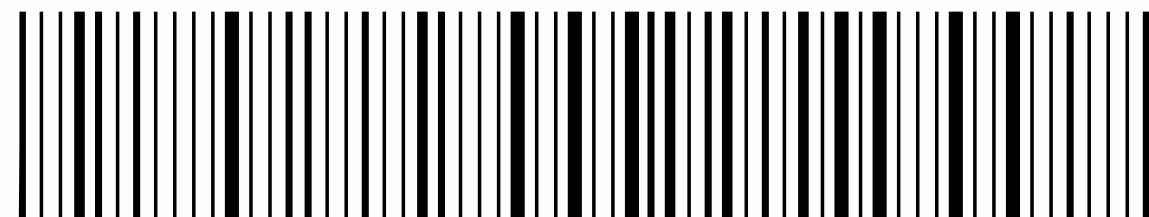


# FLIGHT DELAYS:



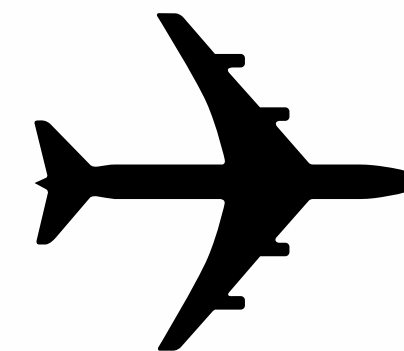
- Is my delayed flight just unlucky?
- Are there any factors that consistently correlates with delays
- Can we choose flights to avoid delays

→ Let a large amounts of historical flight data paint a better picture





# Tools Used:

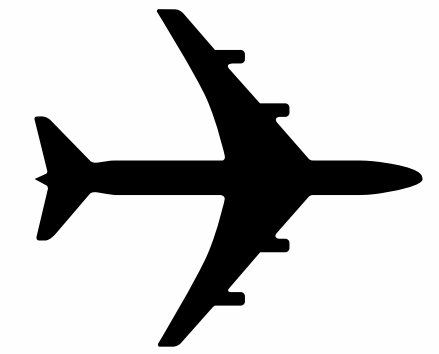


```
library(dplyr)
library(ggplot2)
library(tidyverse)
library(readr)
library(lubridate)
library(ggrepel)
library(jsonlite)
```



# FLIGHT DATA:

## US Bureau of Transportation Statistics



An official website of the United States government [Here's how you know](#)

United States Department of Transportation

### Bureau of Transportation Statistics

Ask-A-Librarian

USA Flight Data from 2024-2025

Data Source: [Transtats](#), Bureau of Transport Statistics, US Department of Transportation

[https://www.transtats.bts.gov/DL\\_SelectFields.aspx?gnoyr\\_VQ=FGJ&QO\\_fu146\\_anzr=b0-gvzr](https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGJ&QO_fu146_anzr=b0-gvzr)

Topics and Geography Statistical Products and Data National Transportation Library Newsroom About

#### TranStats

Search this site:

[Advanced Search](#)

#### Resources

- Database Directory
- Glossary
- Upcoming Releases
- Data Release History

#### Data Finder

**By Mode**

- Aviation
- Maritime
- Highway
- Transit
- Rail
- Pipeline
- Bike/Pedestrian
- Other

**By Subject**

- Safety
- Freight Transport
- Passenger Travel
- Infrastructure
- Economic/Financial
- Social/Demographic
- Energy
- Environment

#### On-Time : Reporting Carrier On-Time Performance (1987-present)

Latest Available Data: February 2026 [Databases](#) [Data Tables](#) [Table Contents](#)

[Download Instructions](#)

Filter Geography:  Filter Year:  Filter Period:

**No data available for March 2026**

☐ Prezipped File ☐ % Missing in table ☐ Documentation ☐ Term

| Field Name                                     | Description   | Support Table                    |
|------------------------------------------------|---------------|----------------------------------|
| <b>Time Period</b>                             |               |                                  |
| <input checked="" type="checkbox"/> Year       | Year          |                                  |
| <input type="checkbox"/> Quarter               | Quarter (1-4) | <a href="#">Get Lookup Table</a> |
| <input checked="" type="checkbox"/> Month      | Month         | <a href="#">Get Lookup Table</a> |
| <input checked="" type="checkbox"/> DayofMonth | Day of Month  |                                  |

```
{r}

years <- c("24","25")
months <- c("01","02","03","04","05","06","07","08","09","10","11","12")
dot_csv <- ".csv"

month_year <- expand.grid(m=months,y=years)
file_names <- paste0(month_year$y,month_year$m,dot_csv)

df <- data.frame()
for(i in file_names) {
  df <- bind_rows(df,read_csv(i))
}

new_colnames <- c("year","month","date","day_of_week","carrier","uniq_carrier",
,"flight_number","orig","orig_state","dest","dest_state","sch_dept","real_dept",
,"sch_arrv","real_arrv","cancelled","cancel_code","sch_flight_time",
,"real_flight_time","flight_distance","delay_carrier","delay_weather","delay_nas",
,"delay_security","delay_late_aircraft")
colnames(df) <- new_colnames
flights <- df

print("Dataset size and head")
dim(flights)
head(flights,6)|
```

# Clean Data and Create Variables:

```
{r}

#remove cancelled flights
flights <- flights |> filter(cancelled==0) |> select(-cancelled,-cancel_code
,-uniq_carrier)

#fill NAs
flights[is.na(flights)] <- 0

#format time + create variables
hhmm_to_time <- function(x) { paste0(substr(x,1,2)," ",substr(x,3,4)) |> hm() }
for(i in 1:14) {
  flights[[i]] <- hhmm_to_time(flights[[i]])
}

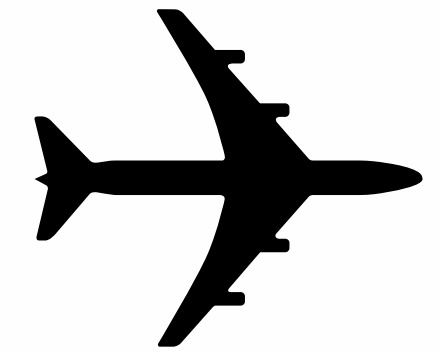
flights <- flights |> mutate(
  dept_delay_min = as.numeric(real_dept - sch_dept, units="mins"),
  arrv_delay_min = as.numeric(real_arrv - sch_arrv, units = "mins"),
  dept_delay_min = ifelse(dept_delay_min < -720, dept_delay_min + 1440,
dept_delay_min),
  arrv_delay_min = ifelse(arrv_delay_min < -720, arrv_delay_min + 1440,
arrv_delay_min),
  dept_delay = ifelse(dept_delay_min > 30,TRUE,FALSE),
  delay_on_air = real_flight_time - sch_flight_time
)
```

Define 'delay' as departing more than 30 minutes behind schedule

| year<br><dbl>            | mo...<br><dbl>      | date<br><dbl>             | day_of_week<br><dbl>    | carrier<br><chr>          | flight_number<br><dbl>       | orig<br><chr>             | orig_state<br><chr> |
|--------------------------|---------------------|---------------------------|-------------------------|---------------------------|------------------------------|---------------------------|---------------------|
| 20...                    | 1                   | 1                         | 1                       | 9E                        | 4814                         | JFK                       | NY                  |
| 20...                    | 1                   | 1                         | 1                       | 9E                        | 4815                         | MSP                       | MN                  |
| dest<br><chr>            | dest_state<br><chr> | sch_dept<br><S4: Period>  |                         | real_dept<br><S4: Period> | sch_arrv<br><S4: Period>     | real_arrv<br><S4: Period> |                     |
| DTW                      | MI                  | 12H 52M 0S                |                         | 12H 47M 0S                | 15H 8M 0S                    | 14H 49M 0S                |                     |
| CLE                      | OH                  | 10H 15M 0S                |                         | 10H 1M 0S                 | 13H 25M 0S                   | 12H 55M 0S                |                     |
| sch_flight_time<br><dbl> |                     | real_flight_time<br><dbl> |                         | flight_distance<br><dbl>  |                              | delay_carrier<br><dbl>    |                     |
| 136                      |                     | 122                       |                         | 509                       |                              | 0                         |                     |
| 130                      |                     | 114                       |                         | 622                       |                              | 0                         |                     |
| delay_weather<br><dbl>   |                     | delay_nas<br><dbl>        | delay_security<br><dbl> |                           | delay_late_aircraft<br><dbl> |                           |                     |
| 0                        |                     | 0                         | 0                       |                           | 0                            |                           |                     |
| 0                        |                     | 0                         | 0                       |                           | 0                            |                           |                     |
| dept_delay_min<br><dbl>  |                     | arrv_delay_min<br><dbl>   |                         | dept_delay<br><lgl>       |                              | delay_on_air<br><dbl>     |                     |
| -5                       |                     | -13                       |                         | FALSE                     |                              | -14                       |                     |
| -14                      |                     | -30                       |                         | FALSE                     |                              | -16                       |                     |
| -4                       |                     | -20                       |                         | FALSE                     |                              | -16                       |                     |
| -7                       |                     | -42                       |                         | FALSE                     |                              | -35                       |                     |
| -5                       |                     | -14                       |                         | FALSE                     |                              | -9                        |                     |
| -7                       |                     | -24                       |                         | FALSE                     |                              | -17                       |                     |
| -8                       |                     | -13                       |                         | FALSE                     |                              | -5                        |                     |
| 5                        |                     | 24                        |                         | FALSE                     |                              | 10                        |                     |

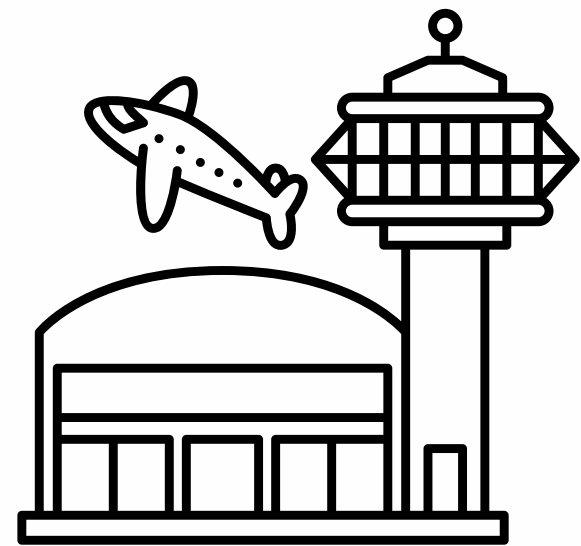


# EDA: How each variable relates with delays



## Semi-Controllable

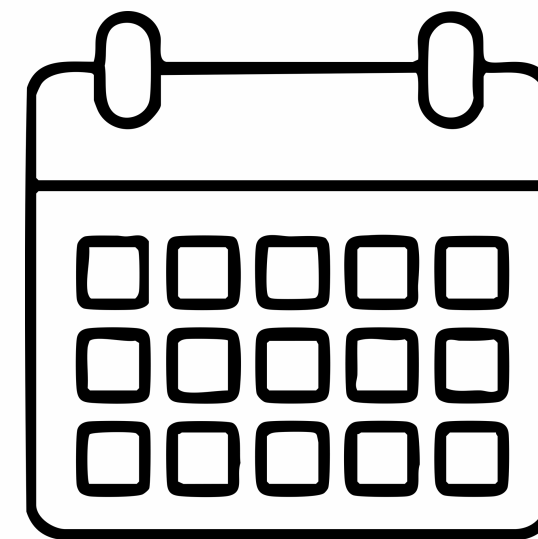
### Uncontrollable



**Airport**



**Flight Distance**



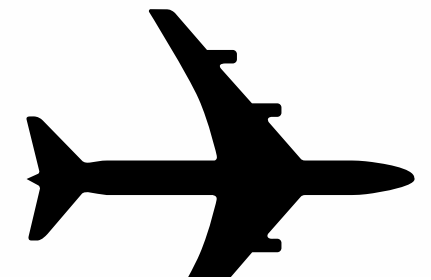
**Flight Date**

### Controllable

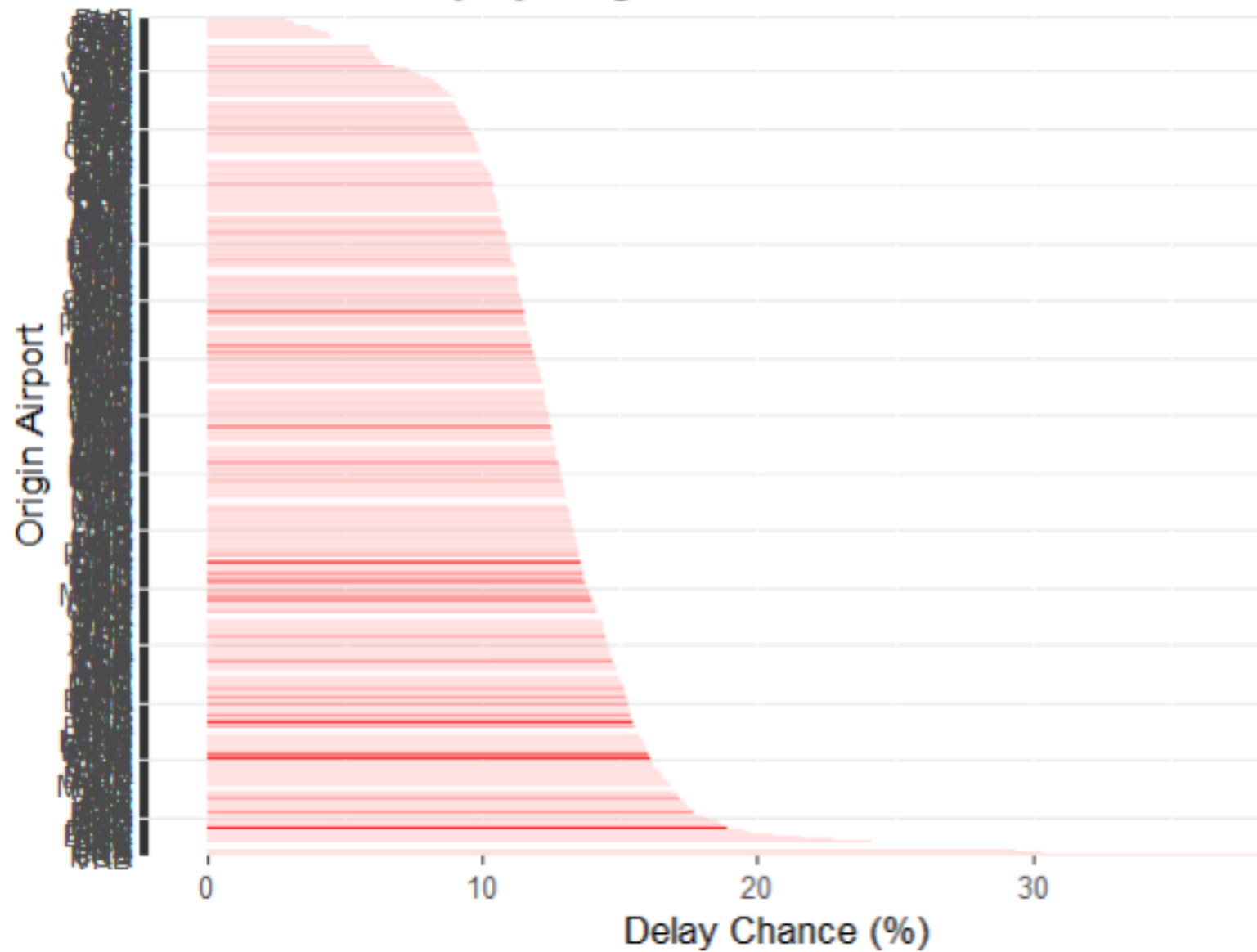


**Airline/Carrier**

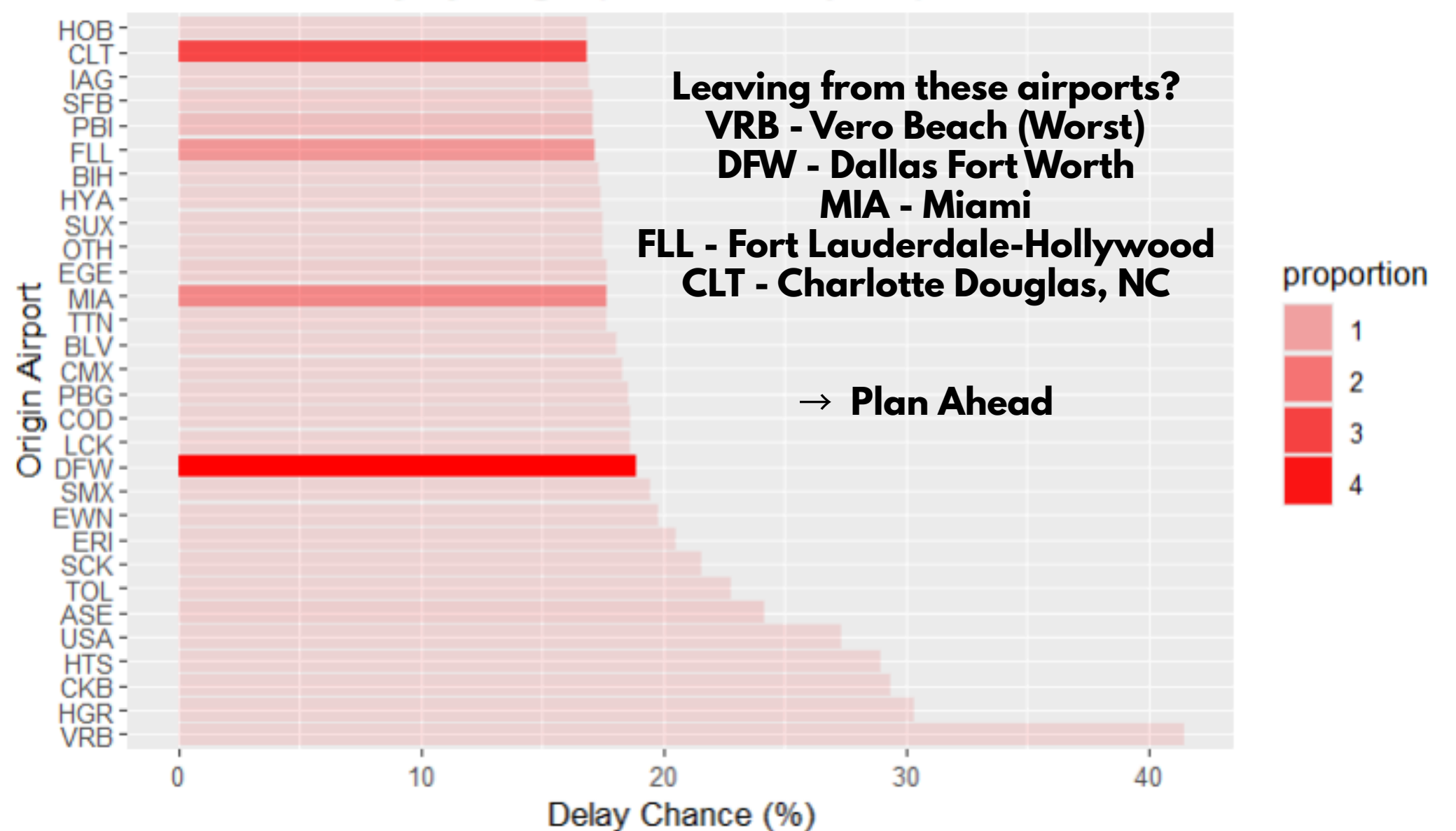
# Does Airport Matter? : Yes



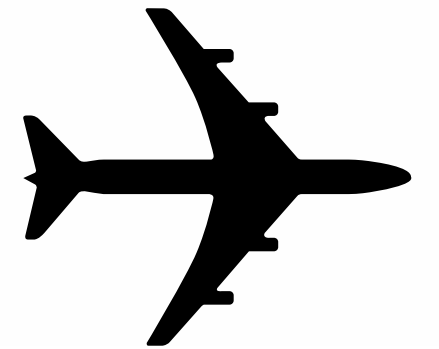
Chance of Delay by Origin



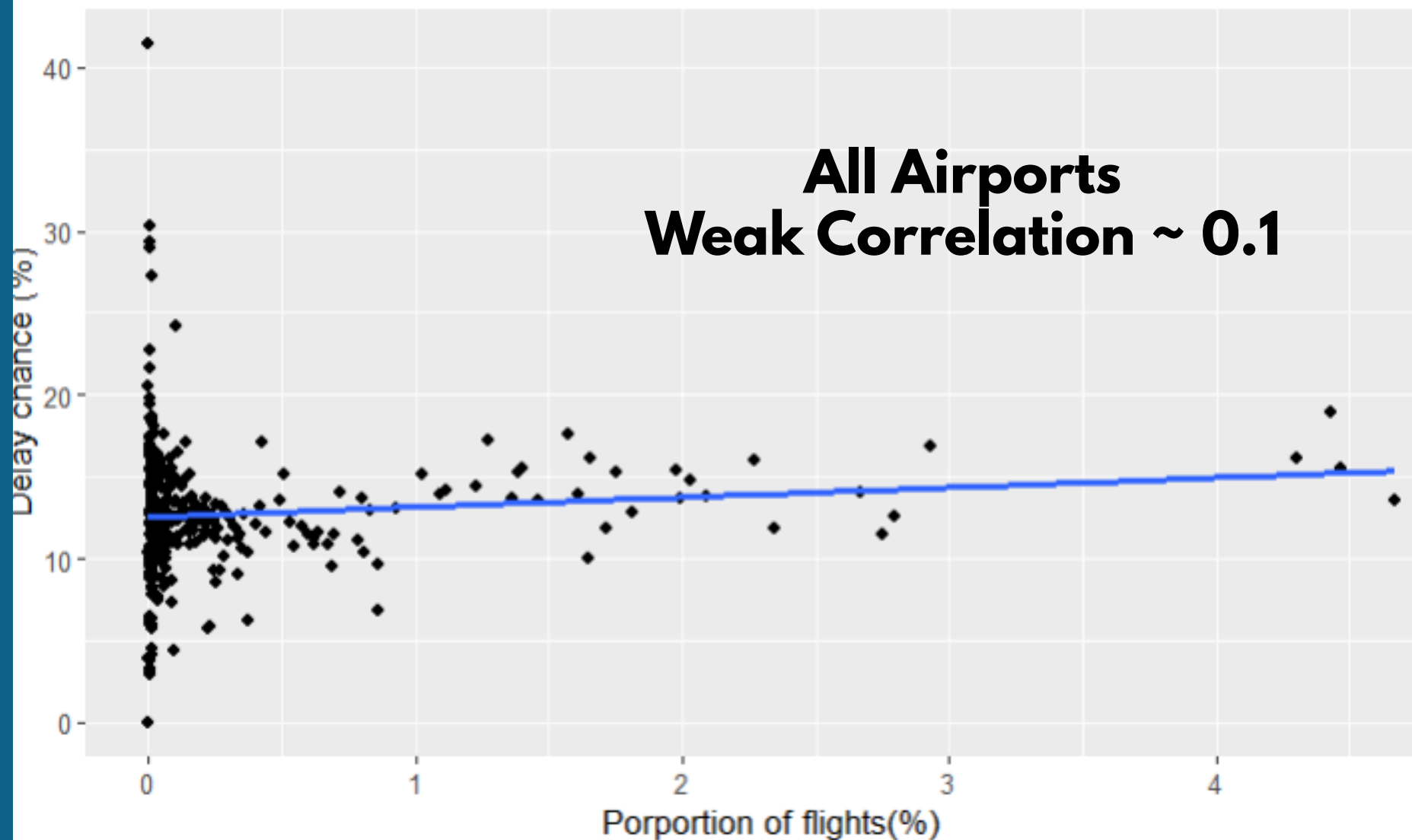
Chance of Delay by Origin (Worst 30 Airports)



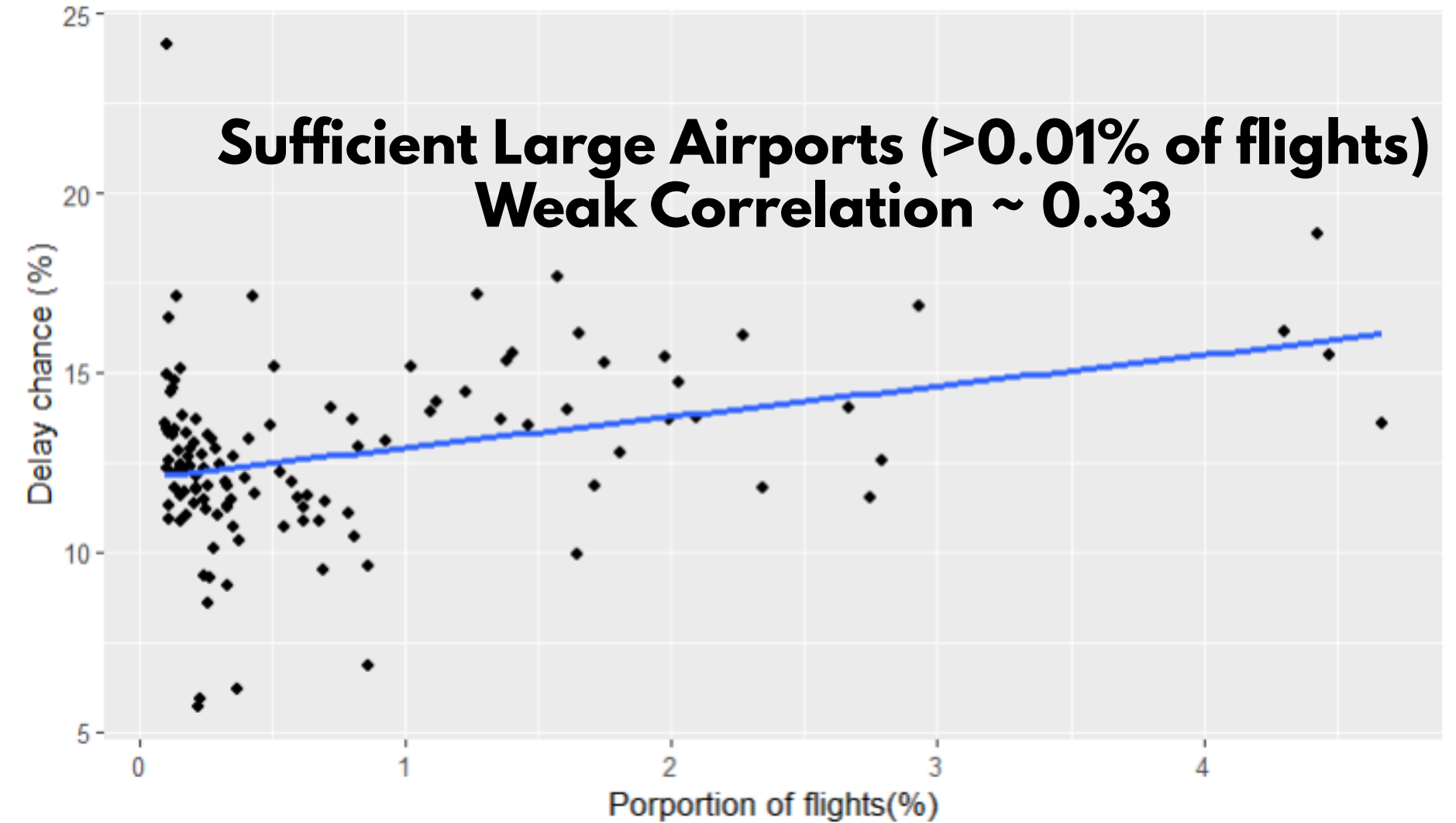
# Major airports vs Regional ones



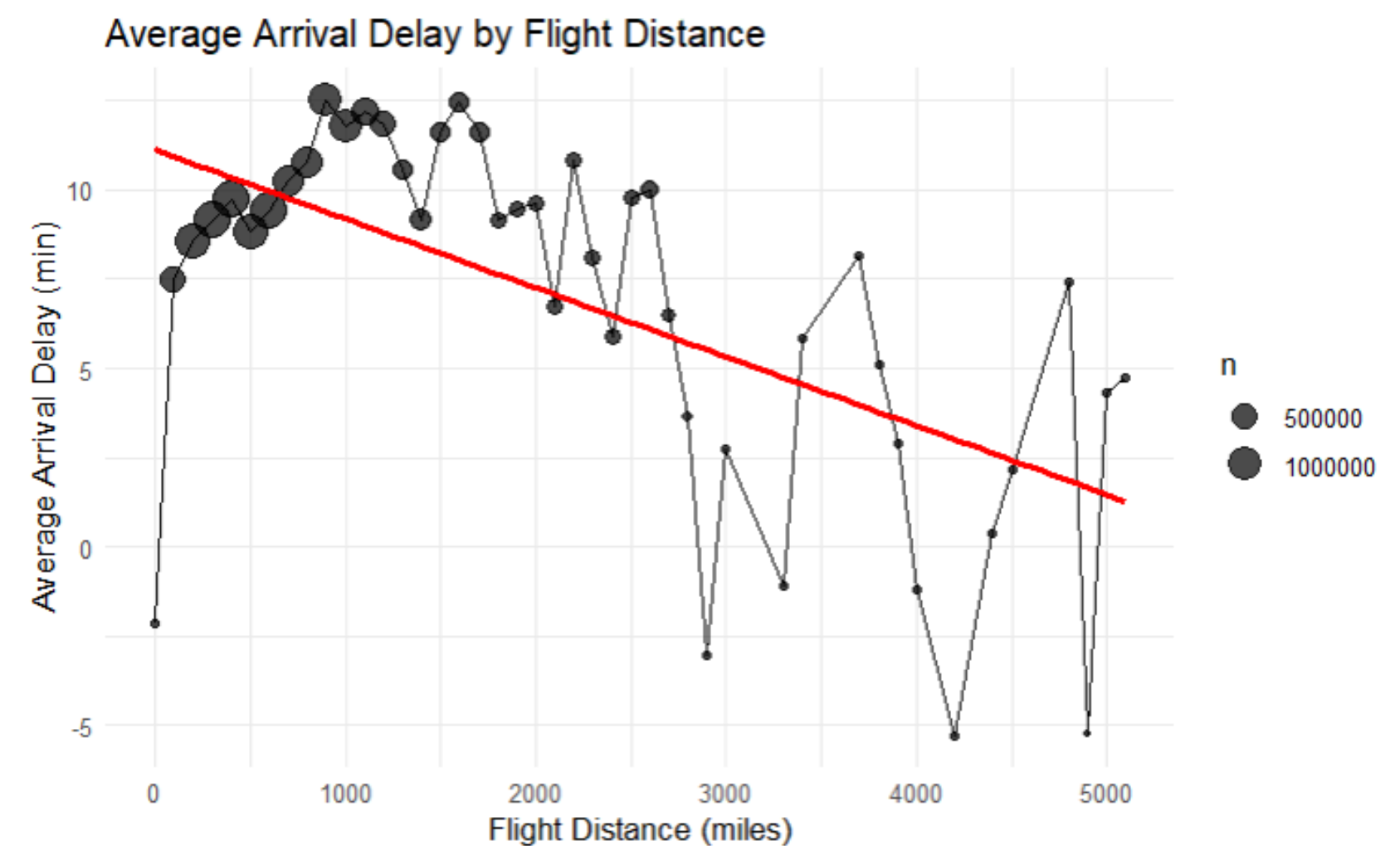
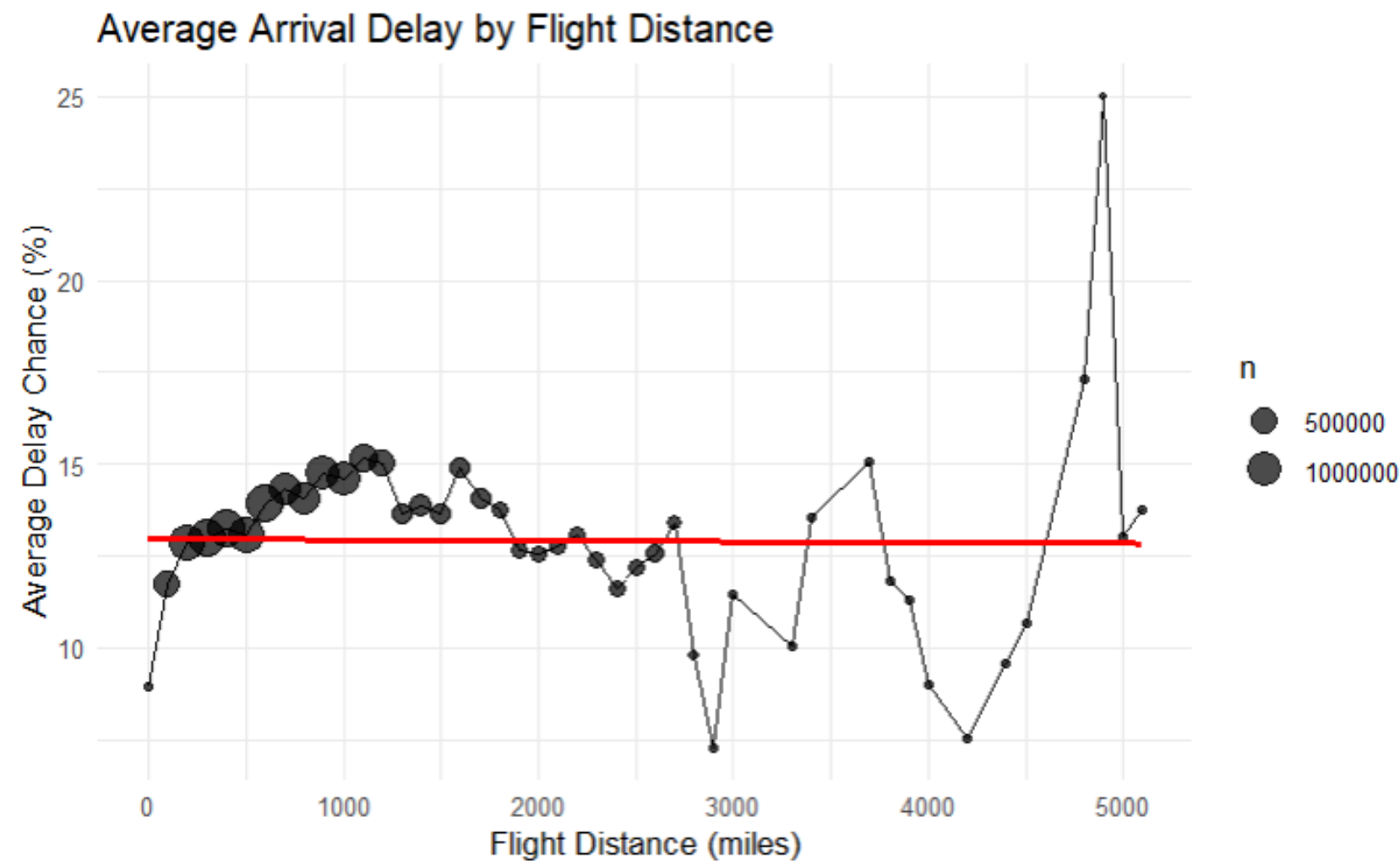
Delay Chance by Airport size



Delay Chance by Airport size

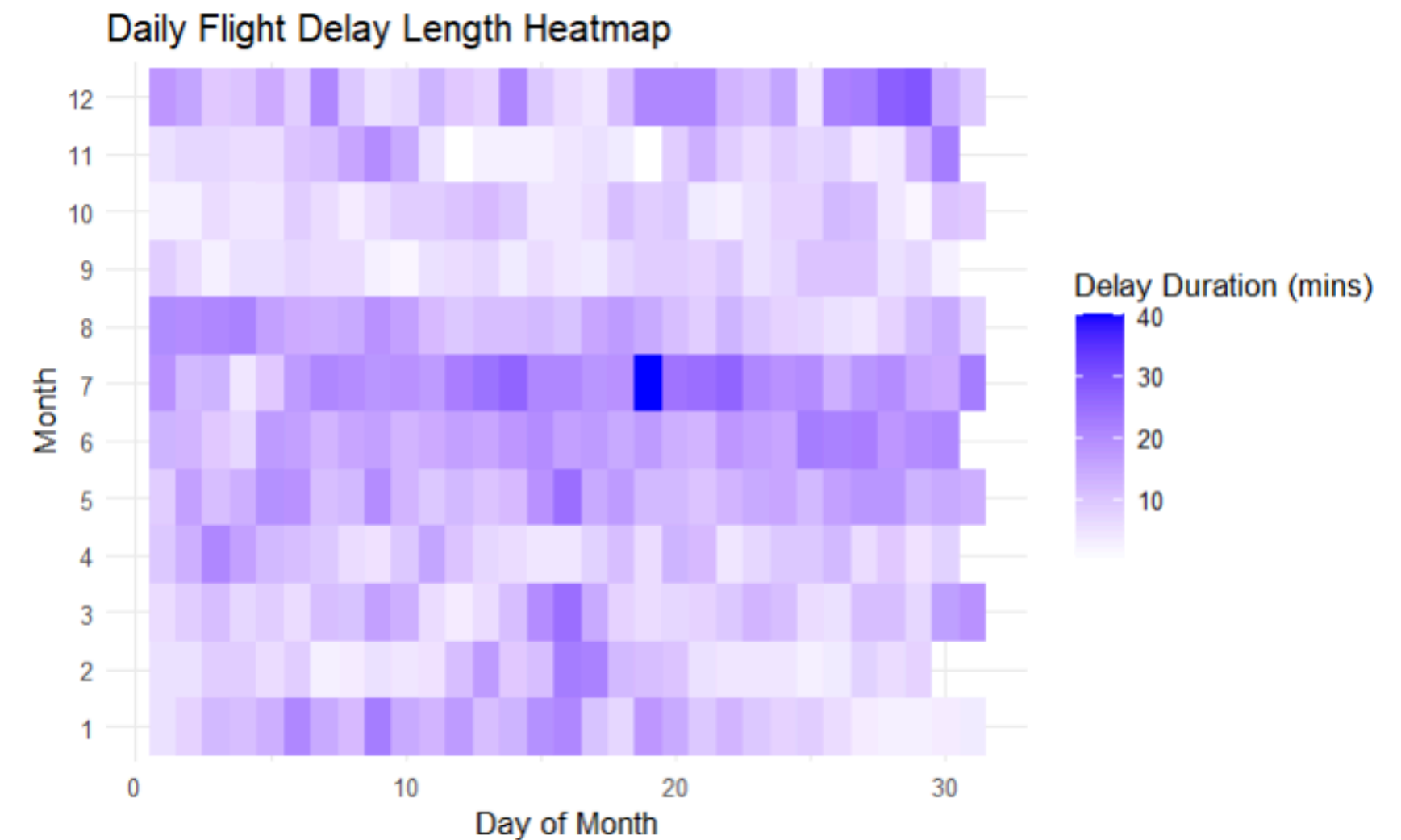
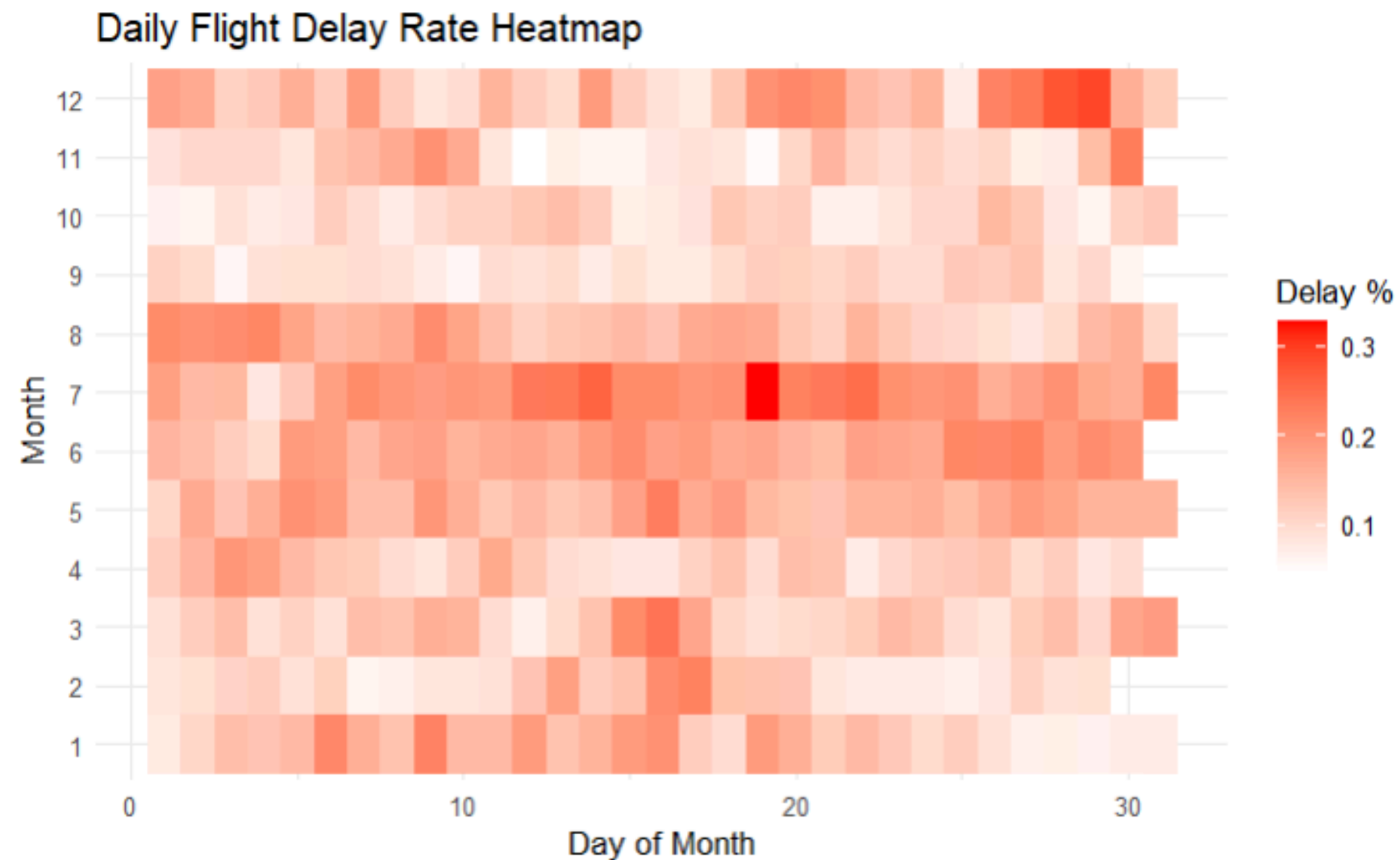


# Longer routes are more delayed?



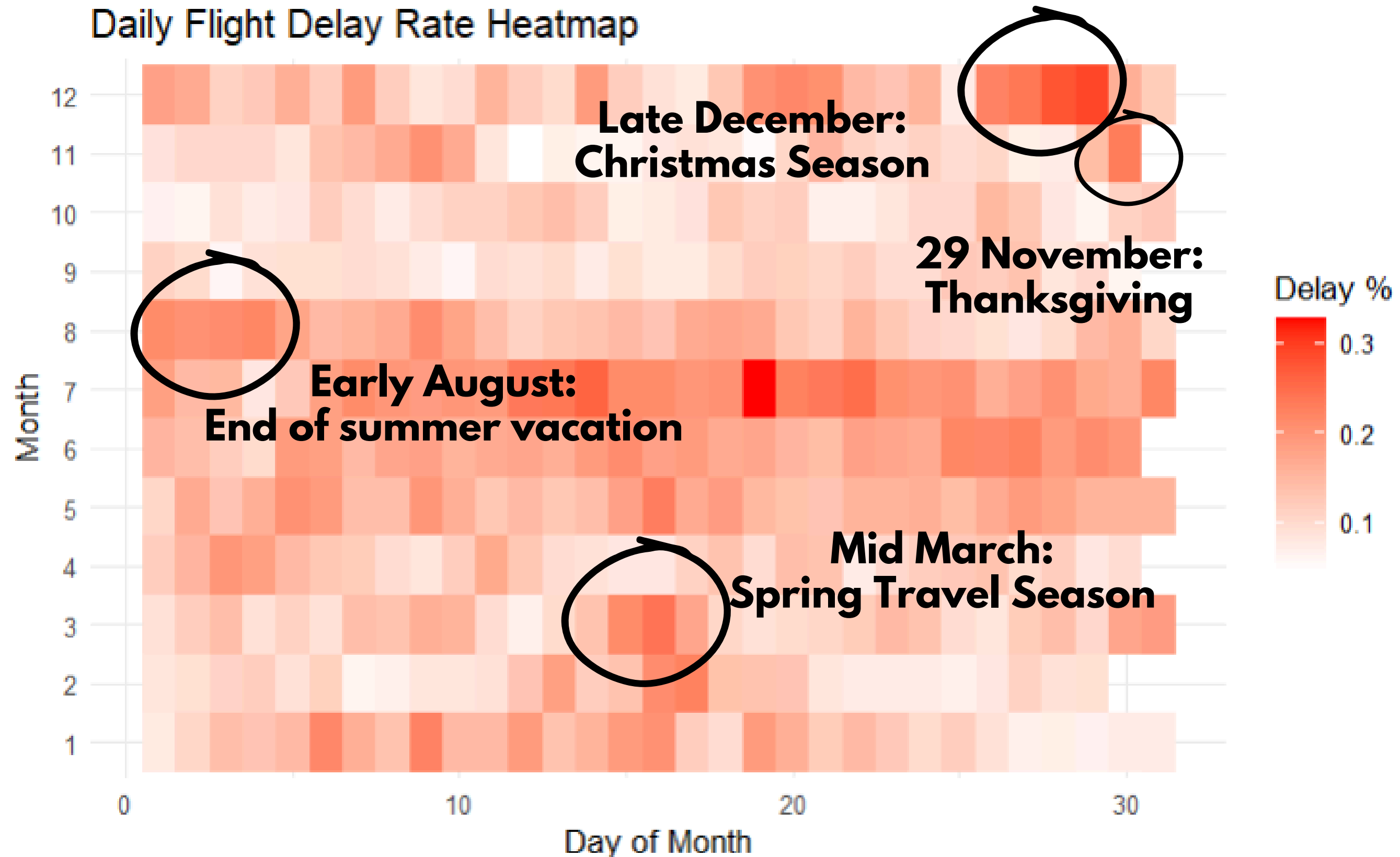
**Route distance has no effect on delay percentage, and are delayed for shorter but negligible times**

# Date: High Seasons cause more delay?



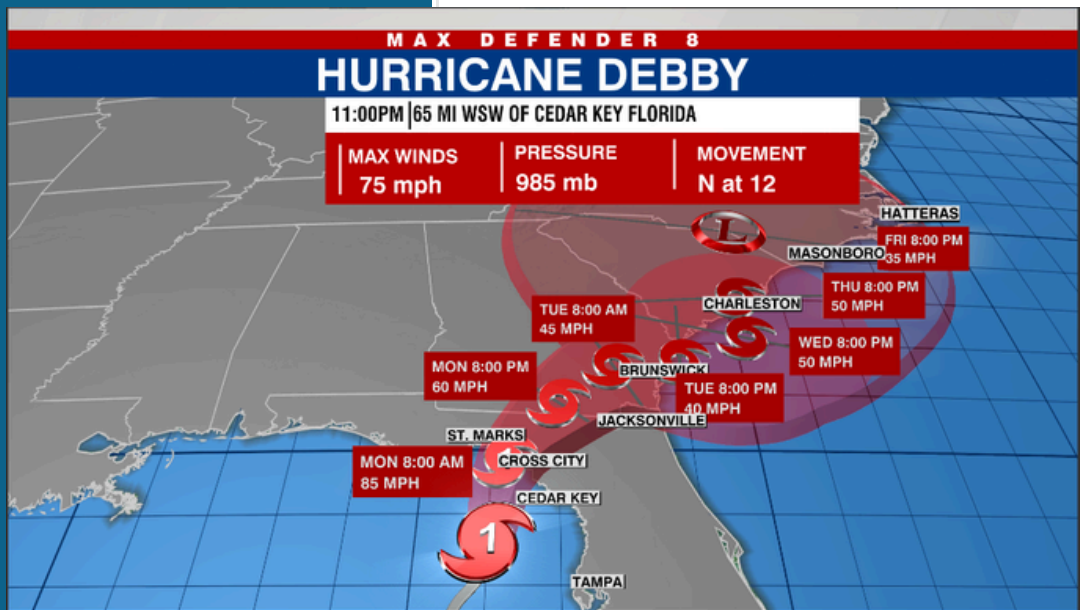


# High Seasons

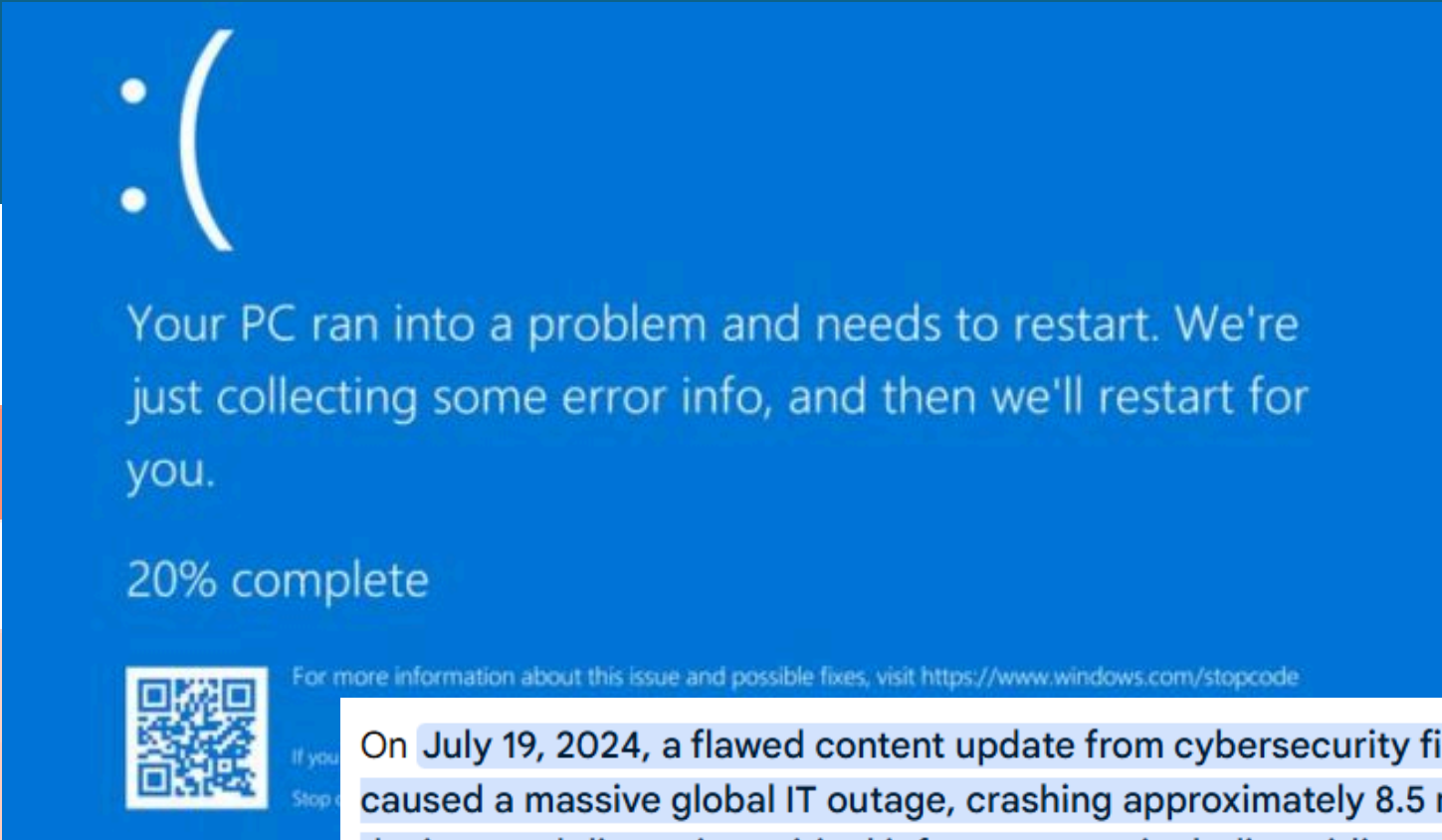




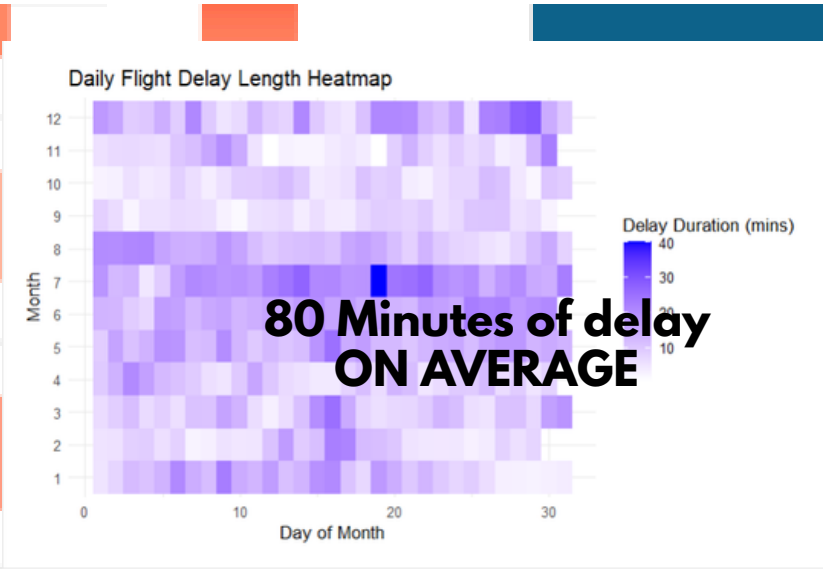
# Incidents



Delay Rate Heatmap

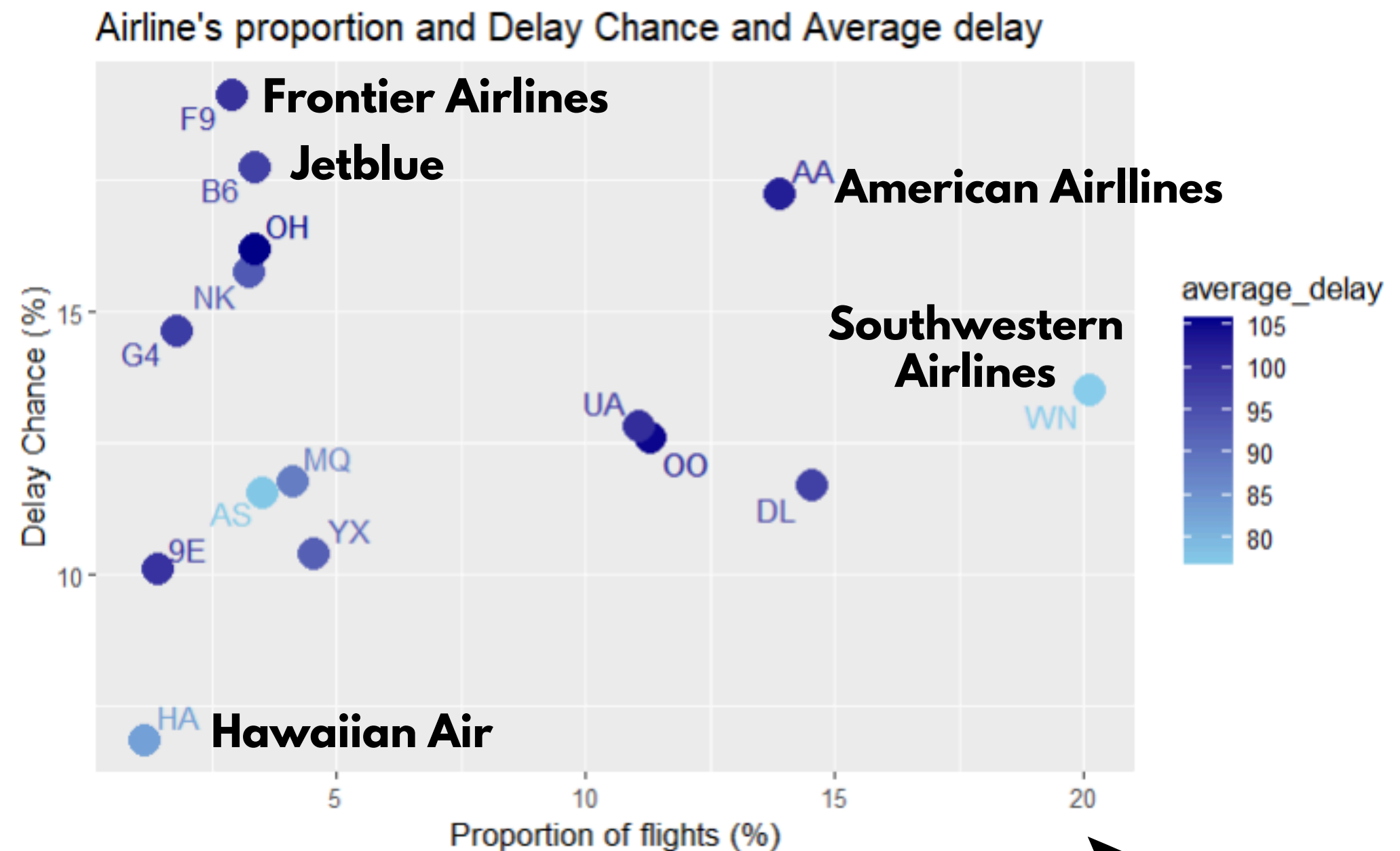


On July 19, 2024, a flawed content update from cybersecurity firm CrowdStrike caused a massive global IT outage, crashing approximately 8.5 million Windows devices and disrupting critical infrastructure, including airlines, hospitals, and banks. The issue was a "null pointer error" in a Falcon sensor update, leading to widespread "blue screen of death" (BSOD) boot loops. [YouTube · LMG Security +2](#)



# Airlines Comparison

Some airlines delay longer and more often than others - a factor when booking flights

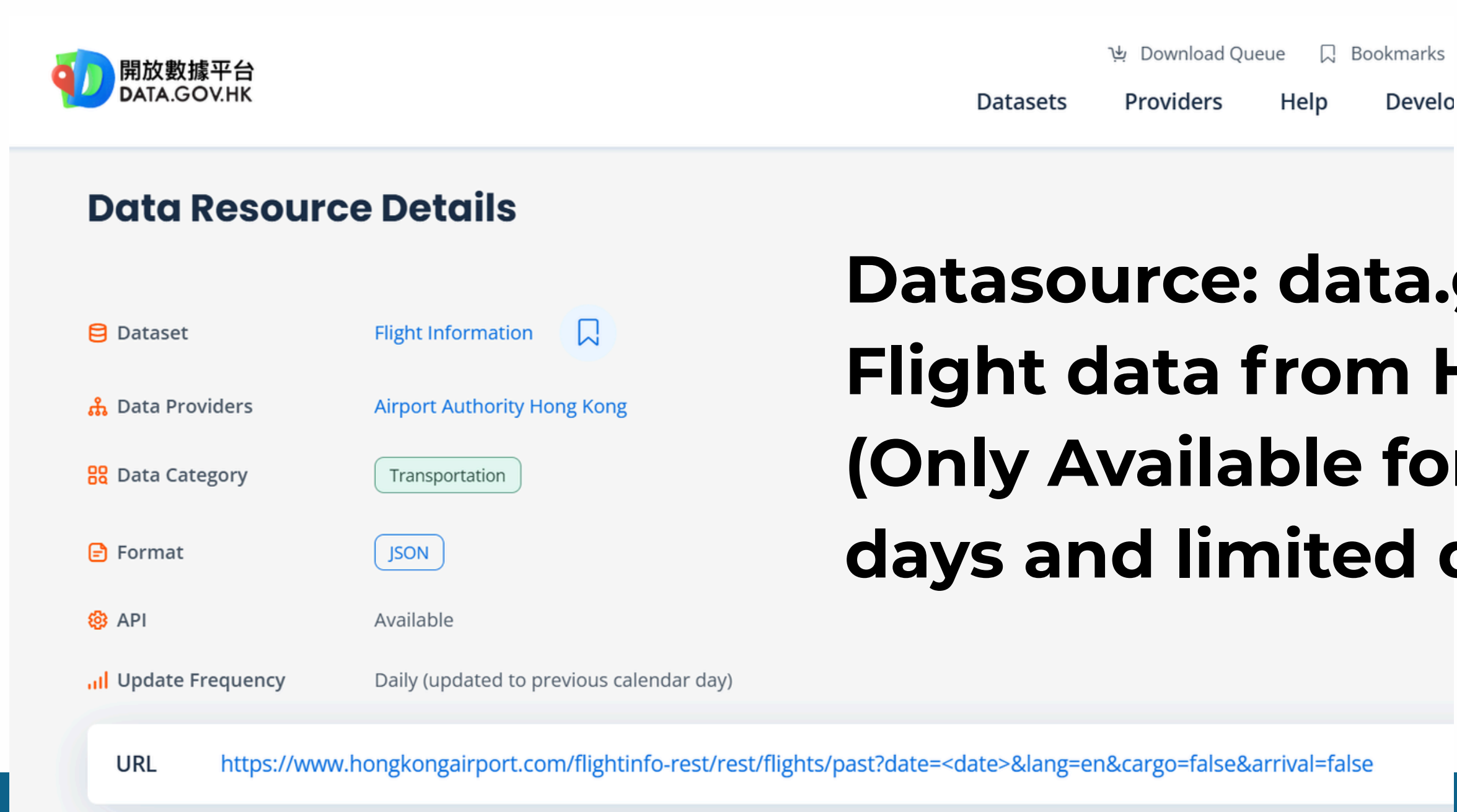


Large Airlines don't delay more or less





# Back to my first question: Is my flight delay just unlucky?



The screenshot shows the Data.GOV.HK website interface. At the top left is the logo with the text '開放數據平台 DATA.GOV.HK'. To the right are links for 'Download Queue' and 'Bookmarks'. Below these are navigation links: 'Datasets', 'Providers', 'Help', and 'Develo'. The main content area is titled 'Data Resource Details' and lists the following information:

- Dataset:** Flight Information (with a bookmark icon)
- Data Providers:** Airport Authority Hong Kong
- Data Category:** Transportation
- Format:** JSON
- API:** Available
- Update Frequency:** Daily (updated to previous calendar day)

At the bottom, the 'URL' is listed as: <https://www.hongkongairport.com/flightinfo-rest/rest/flights/past?date=<date>&lang=en&cargo=false&arrival=false>

**Datasource: data.gov.hk**  
**Flight data from HKG**  
**(Only Available for 90 previous days and limited data features)**

# HKG Flights for March+April 2026

## Scraped using API requests

```
dates_MarApr2026 <- seq(
  from = as.Date("2026-03-01"),
  to = as.Date("2026-04-30"),
  by = "day"
)

get_flights <- function(d){
  url <- paste0(
    "https://www.hongkongairport.com/flightinfo-rest/rest/flights/past?date=",
    d,
    "&lang=en&cargo=false&arrival=false"
  )
  tryCatch(
    raw <- fromJSON(url)$list[[1]] |> as_tibble(),
    error = function(e) NULL
  )

  raw <- raw |> select(time,flight,status,destination) |>
    filter(status!="Cancelled") |>
    unnest(destination) |>
    mutate(sch_dept = hm(time),
           real_dept = hm(substr(status,5,9)),
           delay_min = as.numeric(real_dept - sch_dept, unit="mins"),
           delay_min = ifelse(delay_min < -720, delay_min + 1440, delay_min),
           delayed = ifelse(delay_min>30,TRUE,FALSE),
           date = substr(d,9,10),
           month = substr(d,6,7))
  return(raw)
}

flights_hk <- map(dates_MarApr2026, get_flights)
flights_hk <- bind_rows(flights_hk)
flights_hk_unnested <- flights_hk |> unnest(flight)

flights_hk
flights_hk_unnested
```

A tibble: 27,060 × 10

| time<br><chr> | flight<br><list>     | status<br><chr> | destination<br><chr> | sch_dept<br><S4: Period> |
|---------------|----------------------|-----------------|----------------------|--------------------------|
| 00:05         | <data.frame [1 × 2]> | Dep 00:07       | HND                  | 5M 0S                    |
| 00:05         | <data.frame [1 × 2]> | Dep 00:06       | ICN                  | 5M 0S                    |
| 00:25         | <data.frame [1 × 2]> | Dep 02:50       | LHR                  | 25M 0S                   |
| 00:25         | <data.frame [1 × 2]> | Dep 01:18       | CDG                  | 25M 0S                   |
| 00:25         | <data.frame [2 × 2]> | Dep 00:32       | FRA                  | 25M 0S                   |
| 00:30         | <data.frame [4 × 2]> | Dep 00:34       | MEL                  | 30M 0S                   |
| 00:30         | <data.frame [2 × 2]> | Dep 03:38       | JNB                  | 30M 0S                   |

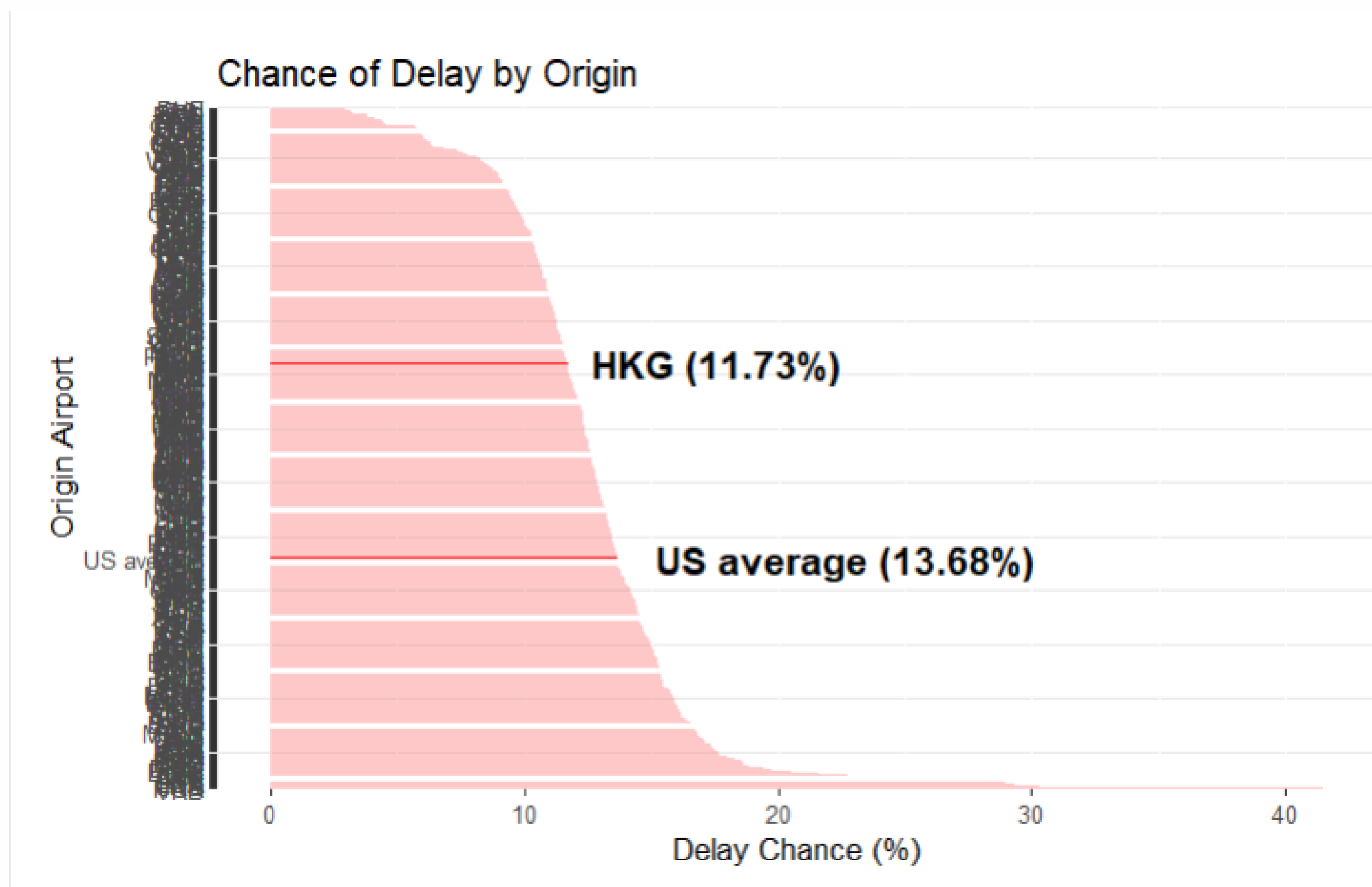
A tibble: 58,015 × 11

| time<br><chr> | no<br><chr> | airline<br><chr> | status<br><chr> | destination<br><chr> | sch_dept<br><S4: Period> | real_dept<br><S4: Period> |
|---------------|-------------|------------------|-----------------|----------------------|--------------------------|---------------------------|
| 00:05         | NH 814      | ANA              | Dep 00:07       | HND                  | 5M 0S                    | 7M 0S                     |
| 00:05         | TW 644      | TWB              | Dep 00:06       | ICN                  | 5M 0S                    | 6M 0S                     |
| 00:25         | CX 237      | CPA              | Dep 02:50       | LHR                  | 25M 0S                   | 2H 50M 0S                 |
| 00:25         | CX 261      | CPA              | Dep 01:18       | CDG                  | 25M 0S                   | 1H 18M 0S                 |
| 00:25         | CX 289      | CPA              | Dep 00:32       | FRA                  | 25M 0S                   | 32M 0S                    |
| 00:25         | LH 7015     | DLH              | Dep 00:32       | FRA                  | 25M 0S                   | 32M 0S                    |
| 00:30         | CX 105      | CPA              | Dep 00:34       | MEL                  | 30M 0S                   | 34M 0S                    |
| 00:30         | AY 5095     | FIN              | Dep 00:34       | MEL                  | 30M 0S                   | 34M 0S                    |
| 00:30         | BA 4567     | BAW              | Dep 00:34       | MEL                  | 30M 0S                   | 34M 0S                    |
| 00:30         | OM 5605     | MGL              | Dep 00:34       | MEL                  | 30M 0S                   | 34M 0S                    |

1-10 of 58,015 rows | 1-7 of 11 columns

Previous **1** 2 3 4 5 6 ... 100 Next

# Airport performance: HKG



# Was that particularly flight being delay just unlucky?

Quite YES

because its on the worse end of the tail

Flights to PVG

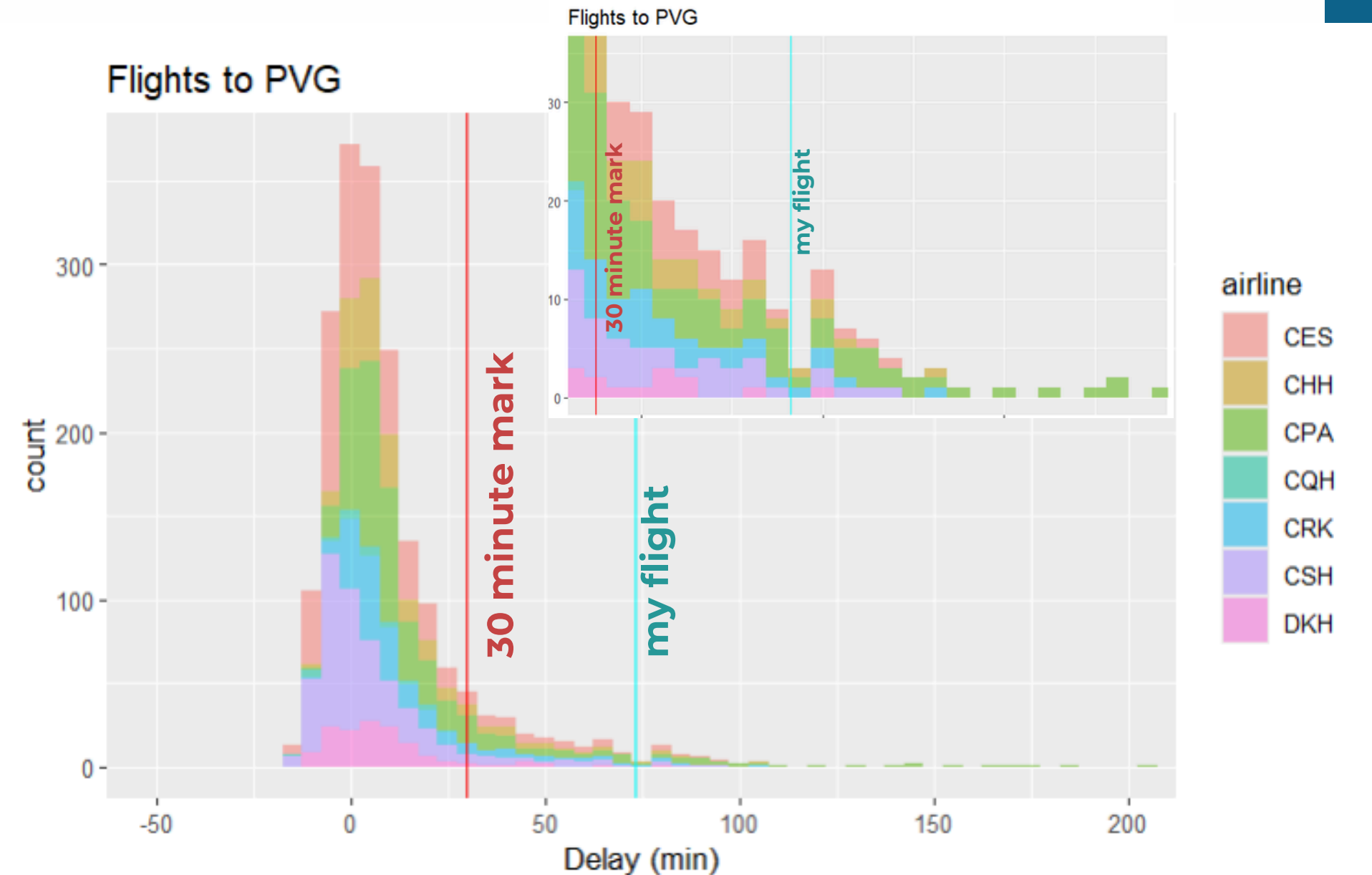
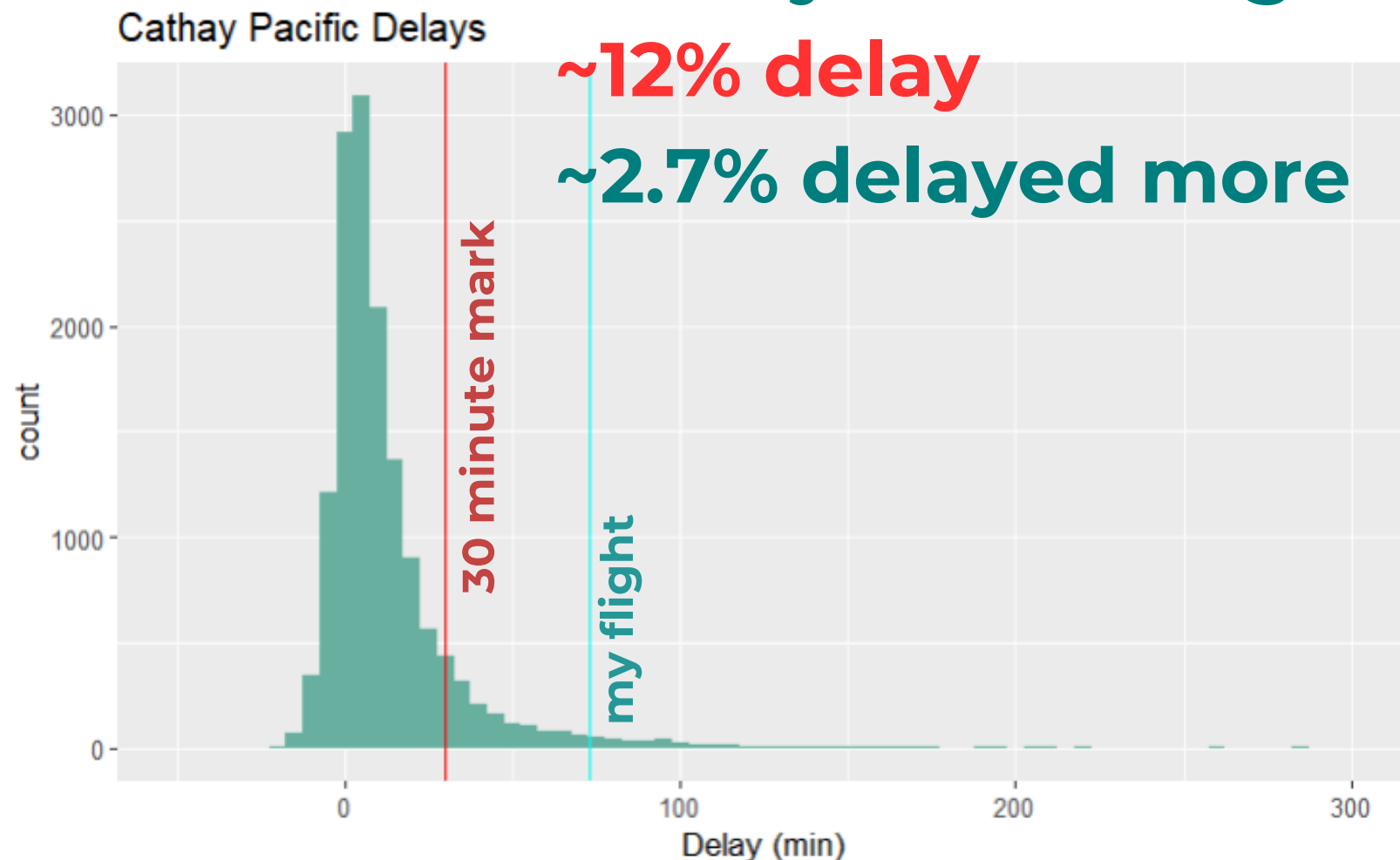
~11% delay

~2.3% delayed more

Cathay Pacific Flights

~12% delay

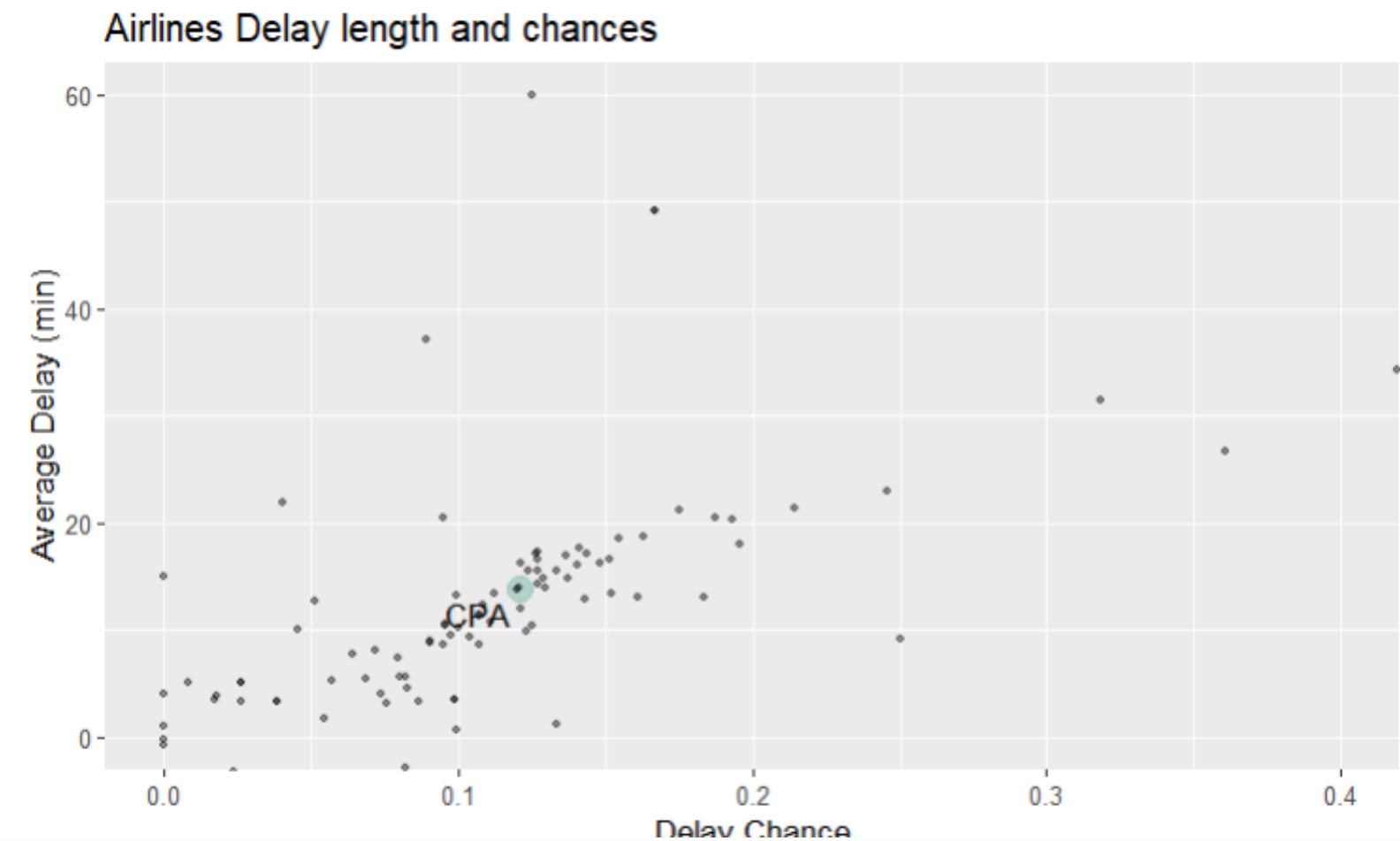
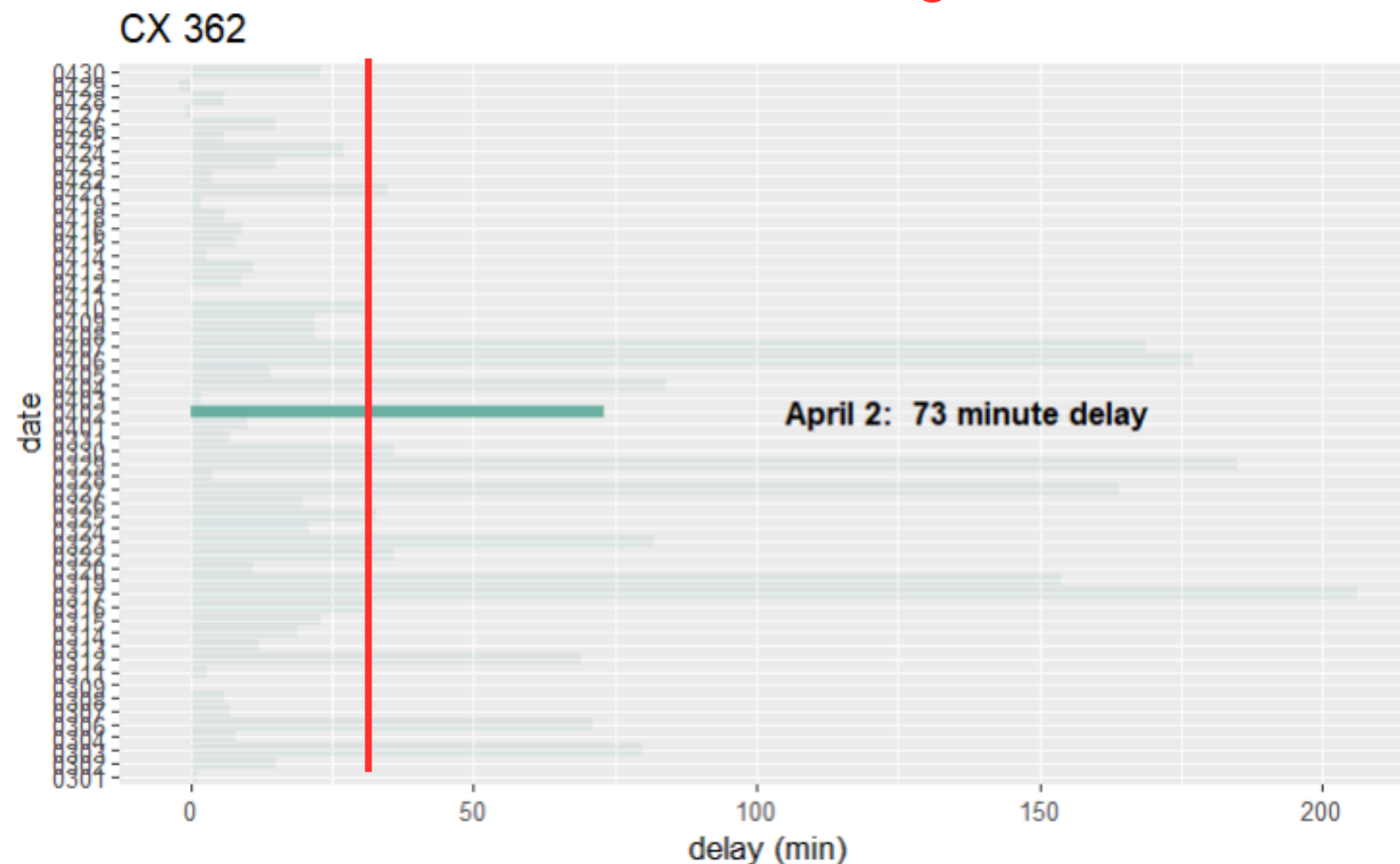
~2.7% delayed more



# Is there something I can do better to avoid it: Probably also a Yes :)

Cathay Flights ~11% delay

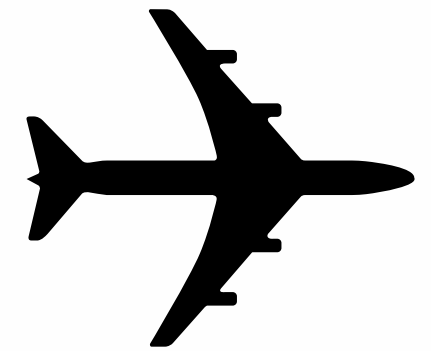
**CX 362 ~23% delay ?!**



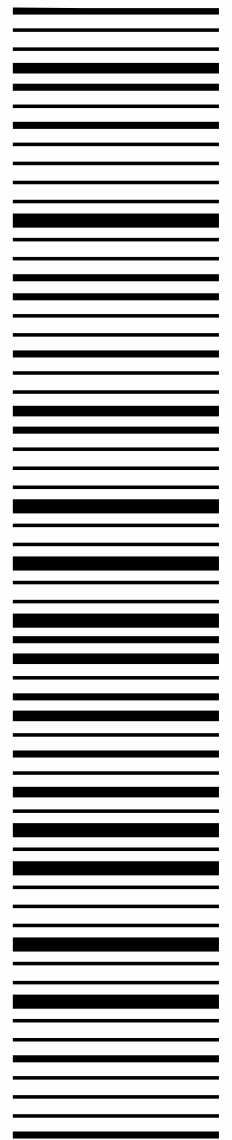
There are airlines that statistically delay less than Cathay Pacific (CPA)



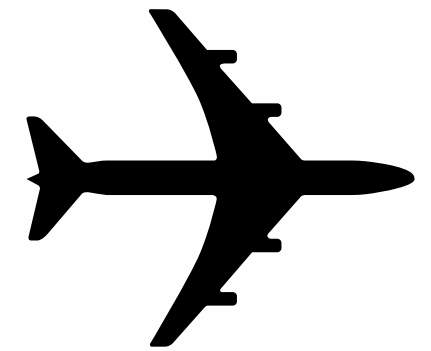
# Key Takeaways



- **Some airports tend to see more delays, its uncontrollable but plan around it.**
- **Hoilday/Peak seasons tend to see more flight delays**
- **Airlines and Specific flights have a variety of delay chances, so if you can't afford a delay (e.g. have a appointment/connecting) choosing can improve your odds**
- **In my case, it was quite unlucky :(**



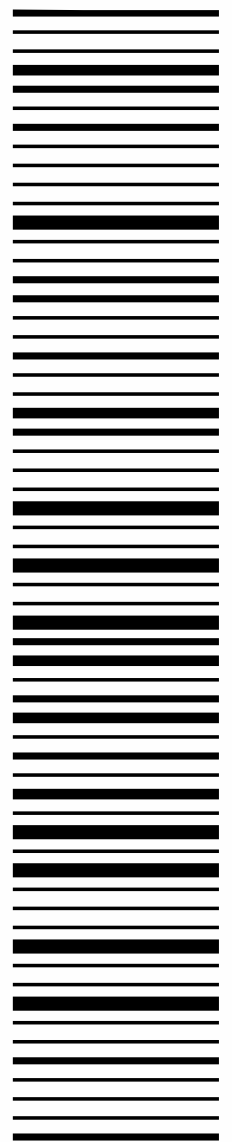
# Acknowledgements



## Data Sources

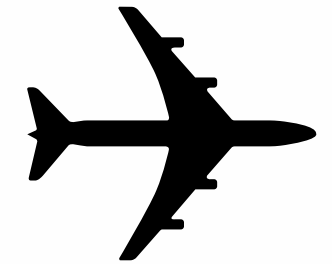
**Hong Kong Government - DATA.GOV.HK API**  
(<https://data.gov.hk/en/>)

**US Bureau of Transportation Statistics**  
(<https://www.bts.gov/>)





THANK YOU



**if:  $\text{cor}(\text{delay}, \text{happiness}) == -1$**   
**do: look at data**  
**else: just don't get unlucky**

---

